

DATA PROCESSING AND EXPERIMENT DESIGN FOR THE RESTORING FORCE SURFACE METHOD, PART I: INTEGRATION AND DIFFERENTIATION OF MEASURED TIME DATA

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The restoring force methods of system identification (which include the Masri/Caughey procedure), require the simultaneous measurement of acceleration, velocity and displacement data. A practical approach requires that one should measure one of these quantities and estimate the other two by numerical integration and/or differentiation. Various methods for achieving this are considered and their relative merits and demerits are assessed.

1. INTRODUCTION

Restoring force methods used for the non-parametric identification of non-linear systems from time data have been attracting a steadily growing measure of attention in recent years. The basic method was initially developed by Masri and Caughey [1] and independently by Crawley and O'Donnell [2]. The simplest form of the procedure allows the non-linear restoring force in a sdof system to be displayed as a surface over the phase plane. Extensions to mdof systems are straightforward [3]. One problem with the method is that the displacement, velocity and acceleration must be obtained simultaneously at each sampling instant. A pragmatic approach to the procedure demands that only one signal should be measured and the other two should be estimated from it. This paper is concerned with determining which signal should be measured and which numerical integration and/or differentiation procedures should be used to determine the other two. As the work progressed, a similar study by Stephens and Yao [4] was brought to the attention of the author. Their paper describes various time-domain methods of data processing for earthquake acceleration records.

Section 2 is a brief discussion of the restoring force approach to identification. The forms of data required are specified. A number of simple time domain methods of integration are described in section 3, the high- and low-frequency problems which can arise are described. Frequency domain integration is the subject of section 4. The fixing of initial conditions i.e. the constants of integration is discussed briefly in section 5. The problem of differentiation of time data is introduced in section 6; specific time and frequency domain differentiators are described in sections 7 and 8 respectively.

2. THE RESTORING FORCE METHOD

For the sake of completeness the identification procedure is outlined here. Considerably more detailed discussions are available elsewhere [1–5]. For the purposes of discussing

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the signal processing aspects of the method, a restriction to sdof systems is perfectly adequate. The extension to mdof systems generates no new problems (except possibly that all measurements from each measurement point on the system should be taken simultaneously at each sampling instant).

The restoring force method is based simply on Newton's second law

$$m\ddot{y} + f(y, \dot{y}) = x(t)$$

where $f(y, \dot{y})$ is the internal restoring force which is a non-linear function of the position and velocity. The mass m and the time histories of the input force x , acceleration $\ddot{y}$, velocity $\dot{y}$ and displacement y are assumed known. The data is usually sampled at regular intervals, spaced Δt apart. The sampled displacements etc. are denoted by $y_k = y(t_k)$, $\dot{y}_k = \dot{y}(t_k)$, ... where t_k is the k th sampling instant given by $t_k = (k-1)\Delta t$.

Given the equation above, one can find the value of the non-linear restoring force f at the sampling instant t_k .

$$f_k = x_k - m\ddot{y}_k.$$

This means that for each sampling instant t_k one has a triplet $(y_k, \dot{y}_k, f_k = f(y_k, \dot{y}_k))$. So for each point in the phase plane $(y_k, \dot{y}_k)$ the value of the restoring force function is known and thus provides one with a point on the restoring force surface $f(y, \dot{y})$.

Given this information one can construct a plot of the restoring force surface using a simple method outlined by O'Donnell and Crawley [2]. One first divides the area of the phase plane covered by the data into small grid squares. If a grid square contains one or more sampled data points $(y_k, \dot{y}_k)$, the force values above the points are averaged to give a force value above the centre of the grid square. This gives a scattering of force values defined on the vertices of a regular grid. One then checks all the empty squares. If an empty square has at least three neighbours containing data, the force values above the neighbour squares are averaged to give a value over the empty one. One then repeats the last step until no new points over grid squares are produced. Because one now has force values defined on a regular grid, the surface (perhaps with holes), is easy to plot. This procedure is very simple but produces a surface which is at best C^1 (i.e. continuous). All force surfaces shown in the figures were generated in this way.

Once one has obtained the $(y_k, \dot{y}_k, f_k)$ triplets one can determine least-squares parameter estimates for a model of the form

$$m\ddot{y} + \sum_{i,j} a_{ij} y^i \dot{y}^j = x(t)$$

or simply

$$m\ddot{y} + c\dot{y} + ky = x(t)$$

if the system is known to be linear. This procedure is described in more detail in [5]. In all cases which follow where a parameter estimation is made, 1000 points of time data are used and a linear least-squares estimator based on the Singular Value Decomposition [6] is used to form the estimates.

The force surface obtained is very sensitive to the mass estimate used, so an accurate estimate is needed before constructing and plotting. If one has no *a priori* knowledge of the mass e.g. from experiment, then one needs to do a least-squares parameter estimation before plotting.

The main problem with the method is that $y(t)$, $\dot{y}(t)$ and $\ddot{y}(t)$ must be measured simultaneously at each sampling instant (and at each measurement point). It would not

be practical to measure all three, particularly if one is dealing with a mdof system. One can circumnavigate this problem in a number of ways, the two most direct ways being:

- (a) measure $\ddot{y}(t)$ and numerically integrate the signal to produce $\dot{y}(t)$ and $y(t)$,
- (b) measure $y(t)$ and numerically differentiate to obtain $\dot{y}(t)$ and $\ddot{y}(t)$.

There are of course other strategies, O'Donnell and Crawley [2] measure displacement and acceleration and then form the velocity using an optimisation scheme. However, the present work is concerned with reporting results obtained in trying to implement procedures (a) and (b) on a digital computer. (Analogue integration has been used with a certain amount of success [7], however it does suffer from most of the problems which arise in the numerical or digital integration.)

The methods of integration considered fall into two categories. (1) Time domain methods i.e. the trapezium rule, Simpson's rule and Tick's rule. (2) Frequency domain methods, i.e. Fourier transform methods. It is assumed throughout that the data is sampled correctly i.e. the sampling frequency is high enough to eliminate aliasing problems. Shannon's theorem demands that one should sample at a frequency at least twice the highest frequency of interest in the data. However, some of the methods applied here require the time signal to vary smoothly from sample point to sample point, for this reason the simulated data is "sampled" at over ten times the highest frequency of interest.

3. TIME DOMAIN INTEGRATION

There are two main problems associated with the integration, the introduction of spurious low frequency components into the integrated signal and the introduction of high frequency components. By using the techniques developed later in section 3.3 one can show that the trapezium rule only suffers from the first of these problems. For this reason the trapezium rule is used in the following discussion.

3.1. LOW FREQUENCY PROBLEMS

In the following studies the data is usually obtained from a computer simulation of the system described by the equation of motion,

$$\ddot{y} + 40\dot{y} + 10^4 y = x(t).$$

For the first simulation, $x(t)$ is a Gaussian white noise sequence $\bar{x} = 0$, $\sigma_x = 1.0$. The system was simulated using a fourth-order Runge-Kutta procedure. The time-step or sampling interval was 0.001 sec, giving a sampling frequency of 1 kHz. Because the Runge-Kutta algorithm becomes unstable at high frequencies, a low-pass Butterworth filter was used on the input to produce a signal in the range 0–200 Hz. The undamped natural frequency of the system above is 100 rad/sec or 15.92 Hz. This means that the sampling is carried out at approximately 30 times the highest frequency of interest. Clearly aliasing is not a problem.

As y , $\dot{y}$ and $\ddot{y}$ are all available from the simulation one can compare the result of the integration procedures with the "exact" results. When a least-squares model of the form

$$m\ddot{y} + c\dot{y} + ky + \alpha = x$$

was fitted to the data, the results were (carets denote estimated quantities throughout):

$$\hat{m} = 1.0000000$$

$$\hat{c} = 40.000000$$

$$\hat{k} = 1.0000001 \times 10^4$$

$$\bar{\alpha} = 0.0$$

and the model mean-square error (MSE) was zero. Because of this result one can be confident that in future, any errors in the parameters reflect the accuracy of the integration procedure and not that of the parameter estimation algorithm.

The (normalised) MSE is used throughout. It is defined by

$$\text{MSE}(y) = \frac{100}{N \cdot \sigma_y^2} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

where y_i is an array of measured time data and $\hat{y}_i$ is the value of y_i predicted by the model; σ_y is the standard deviation of the measured data and N is the number of sample points in the data record. This MSE has the following useful property, if the mean value of the sample $\{y_i\}$, $\bar{y}$ is used as a model i.e. $\hat{y}_i = \bar{y}$ for all i , the MSE is 100.0%. i.e.

$$\text{MSE}(y) = \frac{100}{N \cdot \sigma_y^2} \sum_{i=1}^N (y_i - \bar{y})^2 = \frac{100}{\sigma_y^2} \cdot \sigma_y^2 = 100.$$

The force surface obtained from the first simulation is shown in Fig. 1. The flatness of the surface clearly reflects the fact that the system is linear.

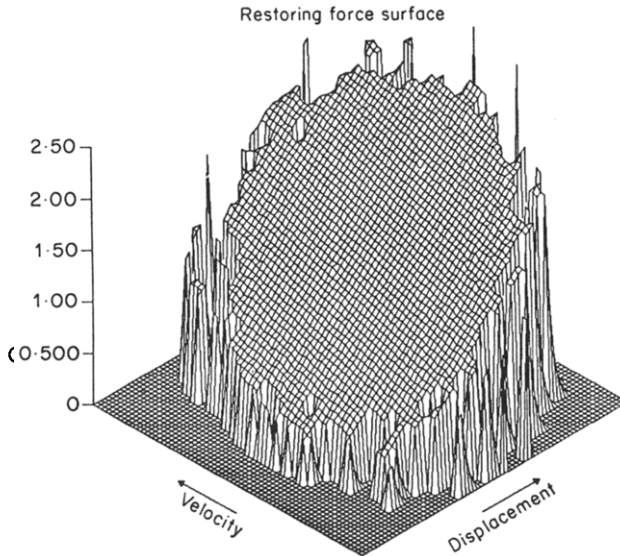


Figure 1. Exact restoring force surface for the simulated linear system used in the examples.

A number of attempts were made to obtain force surfaces by integrating the simulated $\ddot{y}(t)$ data.

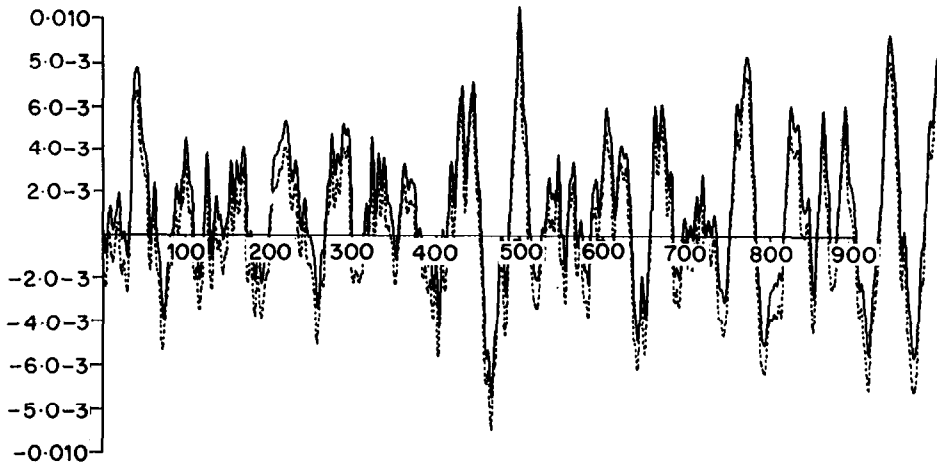
Example 1. The trapezium rule

$$y'_i = y'_{i-1} + \frac{\Delta t}{2} (y_i + y_{i-1}) \quad (1)$$

(where ' indicates the integrated signal) was used to integrate the data $\ddot{y}(t_k)$. Each step introduced a constant of integration

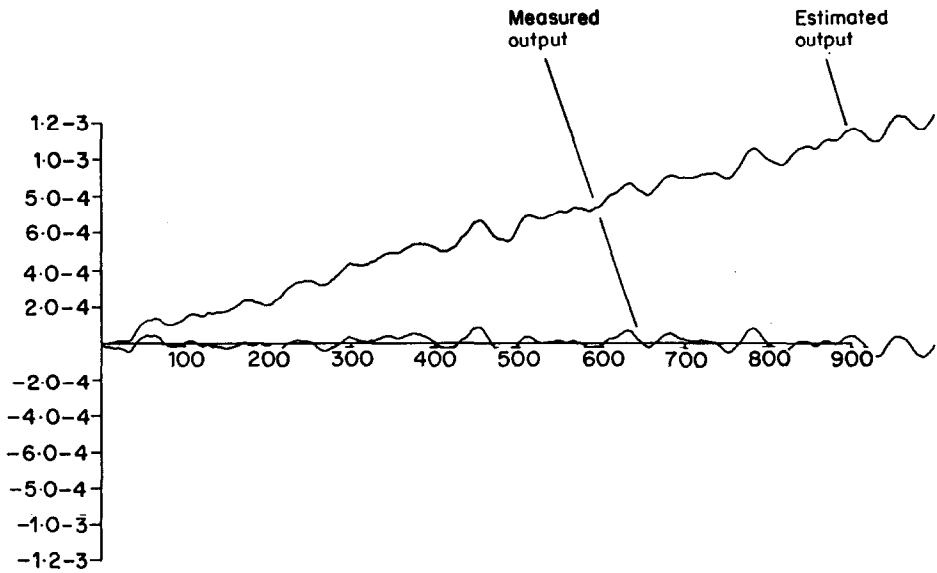
$$\dot{y}(t) = \int dt \ddot{y}(t) + A$$

$$y(t) = \int dt \dot{y}(t) + At + B.$$



Velocity
Normalised MSE: 13.7

Figure 2. Comparison of the exact and estimated velocity data for example 1. ---, Measured output; —, estimated output.



Displacement
Normalised MSE: 4.1+6

Figure 3. Comparison of the exact and estimated displacement data for example 1.

One can clearly see the spurious mean level A in $\dot{y}$ and the linear drift component $At + B$ in y in Figs 2 and 3. Alternatively if one looked at the Fourier transforms (FT) or spectra for $\dot{y}(t)$ and $y(t)$ one would see an unwanted peak in the low frequencies. The $\omega = 0$ or d.c. line is most severely affected. A least-squares parameter fit to the estimated data produced the model

Coefficient	% error
$\hat{m} = 0.773$	22.7
$\hat{c} = 42.9$	7.25
$\hat{k} = 0.968 \times 10^4$	3.2
$\hat{\alpha} = -0.11$	∞

and the model MSE i.e. $MSE(f)$ is 69.8%. The parameter fit is not too bad considering how corrupt the data is. However, plotting the force surface produces the result shown in Fig. 4. It is impossible to infer linearity of the system from this surface.

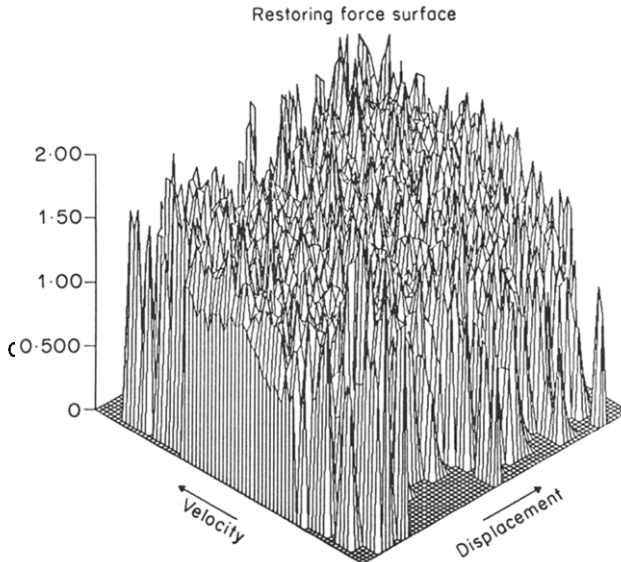
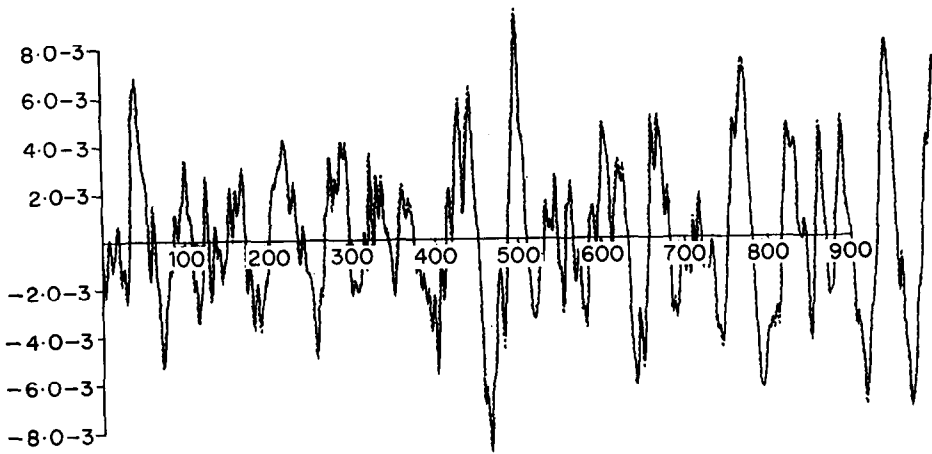


Figure 4. Restoring force surface constructed using the estimated velocity and displacement data obtained in example 1.

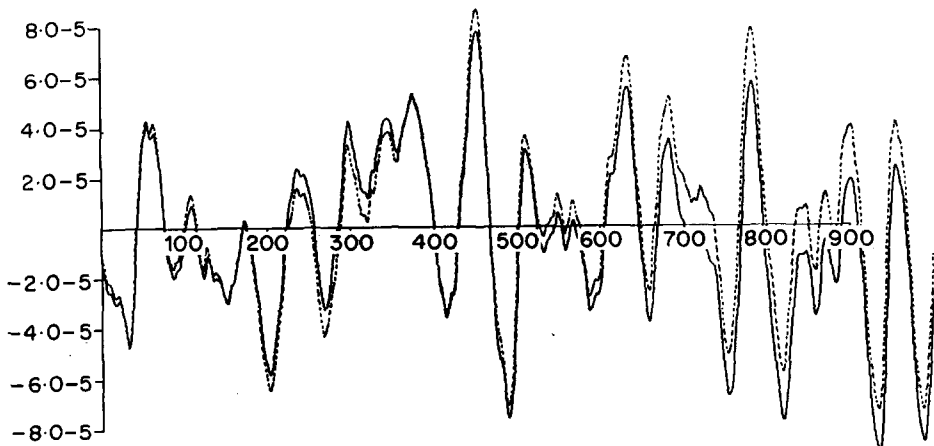
Usually when one integrates a function the constant of integration is fixed by say, initial data $\dot{y}(0) = A$. Unfortunately when one is dealing with a stream of time data one does not have such a condition. However, under certain conditions [$x(t)$ is a zero-mean sequence, $f(y, \dot{y})$ is an odd function of its arguments] one can assume that $\dot{y}(t)$ and $y(t)$ are zero-mean signals. This means that one can set A and B equal to the appropriate values by removing the mean level from $\dot{y}(t)$ and removing a least-squares linear trend from $y(t)$.

Example 2. Again the signal $\ddot{y}(t)$ was integrated twice using the trapezium rule. The mean of the resulting signal $\dot{y}(t)$ was estimated and removed numerically. Similarly, a least-squares straight line was fitted to the $y(t)$ data and removed. The comparison between the exact and estimated signals is shown in Figs 5 and 6. The estimate is much better,



Velocity
Normalised MSE: 0.718

Figure 5. Comparison of the exact and estimated velocity data for example 2. In this case the mean has been removed from the estimated data. ---, Measured output; —, estimated output.



Displacement
Normalised MSE: 12.4

Figure 6. Comparison of the exact and estimated displacement data from example 2. In this case a linear drift component has been removed from the estimated data. ---, Measured output; —, estimated output.

$MSE(\dot{y}) = 0.718$ and $MSE(y) = 12.4$. It is clear that the mean removal is sufficient to fix the velocity estimate to a high order of accuracy, however, higher order polynomial trends remain in the displacement and these must be removed by other methods. The parameter estimates for this data are

Coefficient	% error
$\hat{m} = 0.997$	0.3
$\hat{c} = 40.02$	0.05
$\hat{k} = 0.939 \times 10^4$	6.1
$\hat{\alpha} = 0.068$	∞

and the MSE for the model is 7.18%. The force surface obtained is shown in Fig 7. The force surface and model coefficients are much better now, but the surface still does not compare favourably with the “exact” one. One can improve the results by removing higher order trends from the displacement. Before discussing this, another effect is worth mentioning. When one measures $x(t)$ and $\ddot{y}(t)$ it is possible that a d.c./mean level can be introduced by the instrumentation. Before constructing the force surface this must be removed. It seems reasonable to do this before the integration step. However, this may cause problems. Although $\ddot{y}(t)$ may be a zero-mean signal any finite sample of $\ddot{y}$ will have a non-zero mean value $\ddot{y}_s$, so subtracting this produces a signal $\ddot{y} - \ddot{y}_s$ which is not zero-mean. Integrating this signal, one obtains

$$\dot{y}(t) = \int dt \ddot{y}(t) - \ddot{y}_s t + A$$
$$y(t) = \int dt \dot{y}(t) - \frac{\ddot{y}_s}{2} \cdot t^2 + At + B$$

and it is now necessary to remove a linear trend from $\dot{y}(t)$ and a quadratic trend from $y(t)$. The rather dramatic result of removing the $\ddot{y}$ mean initially on the integrated displacement is shown in Fig. 8.

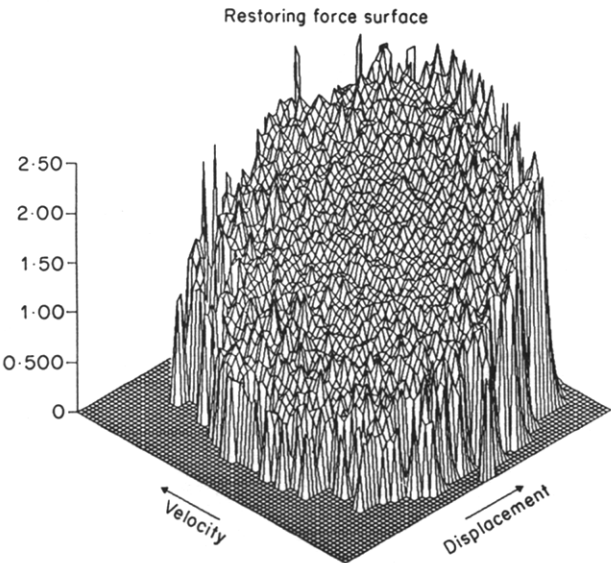


Figure 7. Force surface obtained using the estimated velocity and displacement data from example 2.

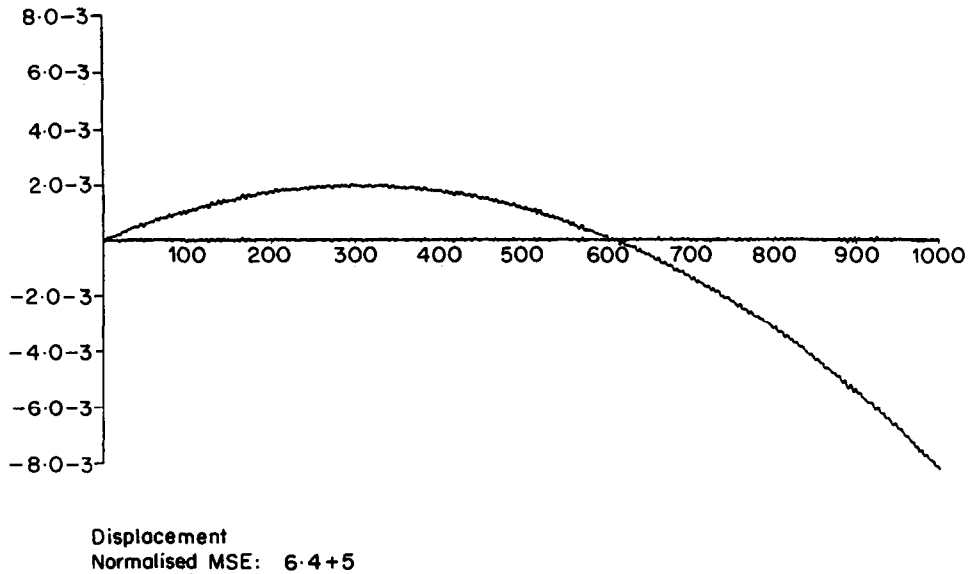


Figure 8. Comparison of the exact displacement data from example 2 with that estimated by integration after the acceleration mean-level was removed. The introduction of a quadratic trend is clearly shown. ---, Measured output; —, estimated output.

It is obvious from the examples above that one is not removing all the low-frequency contamination from the displacement signal. There are essentially two ways of removing low-frequency trends, namely least-squares polynomial trend removal and high-pass filtering.

First, polynomial trend removal is considered. A model of the form

$$y(t) = \sum_{n=1}^{n_{\max}} a_n t^n$$

is fitted over the whole data record and the identified trend is removed from the data. The results of fitting and removing polynomials of various orders are displayed in Table

TABLE 1
Normalised mean-square errors in the estimated velocity and displacement if polynomial trends of various orders are removed

Order	MSE ($\dot{y}$)	MSE (y)
1	0.718	12.353
2	0.718	7.604
3	0.718	5.208
4	0.719	4.969
5	0.719	5.119
6	0.718	4.085
7	0.719	4.080
8	0.718	4.061
9	0.718	4.037
10	0.718	5.653

1. It is clear from the table that there is little point in removing trends of order higher than $n = 6$. In fact, if one removes trends of too high an order, one is also removing low frequency data which should be there. This means that the MSE begins to increase with polynomial order after passing through a (rather shallow) minimum.

The alternative to this is high-pass filtering. Again, a balancing act is required. If the chosen cut-off is too low the data remains corrupted. If the cut-off is too high one removes data which should be there. It is unfortunate, but there appears to be no simple method of distinguishing between the actual data and the low-frequency noise introduced by the integration procedure. Table 2 shows the integration error as a function of cut-off frequency.

TABLE 2
Mean-square errors in the integrated velocity and displacement data after high-pass filtering to remove the low frequency component introduced by the integration

Cut-off (Hz)	MSE ($\dot{y}$)	MSE (y)
0.5	0.700	8.029
1.0	0.709	4.609
2.0	0.715	4.318
3.0	0.811	6.172

Example 3. The data was integrated as before. The velocity and displacement data were high-pass filtered using a digital Butterworth filter with cut-off at 1.0 Hz. The resulting data gave $\text{MSE}(\dot{y}) = 0.709$ and $\text{MSE}(y) = 4.61$. A least-squares fit produced the results

Coefficients	% error
$\hat{m} = 0.974$	2.6
$\hat{c} = 40.6$	1.5
$\hat{k} = 0.899 \times 10^4$	10.1
$\hat{\alpha} = 0.0$	0.0

with a model MSE of 11.2%. It is worth mentioning that as the force surface is linear, if the input force and acceleration had been filtered in the same way as the velocity and displacement data a perfect force surface would possibly have been obtained. This is not the case for a non-linear system as one is essentially assuming that the principle of superposition applies when one makes the above assertion.

It is a fairly simple matter to show that polynomial trend removal is equivalent to high-pass filtering. Suppose one is given a record of time data of length T sec sampled at Δt intervals. If one fits a polynomial of order n to the data, one can account for n zero-crossings in the record. As there are two zero-crossings per cycle, this accounts for $n/2$ cycles. $n/2$ cycles in T sec indicates a signal of frequency $n/(2T)$ or $n/(2N\Delta t)$ where N is the number of points in the record. This means that removing a polynomial trend of order n is equivalent to high-pass filtering with cut-off $n/(2N\Delta t)$.

Filtering can be rather time consuming as the data must be passed through the filter both forwards and backwards in order to zero the phase lag introduced by a generic filter.

The phase lag can manifest itself as a time delay which destroys the simultaneity of the input and output. This can produce serious distortions in the force surface.

3.2. HIGH FREQUENCY PROBLEMS

The other two time domain methods considered here, Simpson's rule and Tick's rule both suffer from the same low-frequency problems as the trapezium rule. Unfortunately they are also unstable at high frequencies. This means that the integrated signals must be band-pass filtered rather than simply high-pass filtered.

Simpson's rule is given by

$$y'_{n+1} = y'_{n-1} + \frac{\Delta t}{3} \cdot (y_{n+1} + 4y_n + y_{n-1}) \quad (2)$$

where the signal y' is the integral of y as before. Tick's rule is given by the slightly more complicated formula

$$y'_{n+1} = y'_{n-1} + \Delta t \cdot (0.3854y_{n+1} + 1.2832y_n + 0.3854y_{n-1}).$$

Although the latter algorithm is slightly better over the low frequencies than Simpson's rule, it is more badly behaved i.e. more unstable at high frequencies.

Example 4. The same system as before was simulated with $x(t)$ a Gaussian white-noise sequence band-limited to 5–40 Hz i.e. the acceleration contained no high frequency part. The data was then integrated using Tick's rule. An enormous high-frequency component was introduced into the signal as shown in Fig. 9.

Simpson's rule is intermediate between the trapezium rule and Tick's rule i.e. it has similar low frequency behaviour and intermediate high frequency behaviour. It was found throughout that for the examples considered, the extra accuracy offered by Simpson's rule was unimportant if the data was sampled at an appropriately high rate.

Example 5. The same signal as in example 4 above was integrated using the trapezium rule. The mean was removed from the estimated velocity and a linear drift component

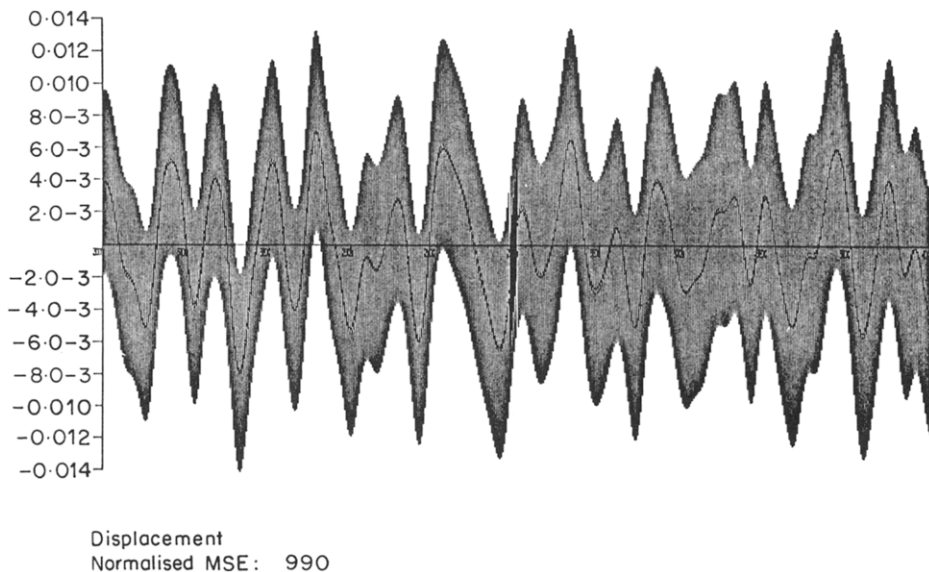


Figure 9. Comparison of exact and estimated displacement data showing the large high-frequency component introduced into the estimate if Tick's rule is used. ---, Measured output; —, estimated output.

was removed from the estimated displacement. The resulting errors were $MSE(\dot{y}) = 0.159$ and $MSE(y) = 0.197$. An LS parameter estimation yielded the results

Coefficients	% error
$\hat{m} = 1.02$	2.0
$\hat{c} = 40.6$	1.5
$\hat{k} = 1.03 \times 10^4$	3.0
$\hat{\alpha} = 0.00$	0.0

and a model MSE of 0.08%. The force surface obtained is shown in Fig. 10. The results are excellent. This example clearly suggests that, rather than using more complicated data processing, one should choose the excitation for the system in such a way as to minimise processing problems. After filtering the input, there appears to be so little energy in the output at low frequencies that the integration procedure does not magnify it to the extent that it becomes a problem. These ideas will be considered in more detail in the following paper.

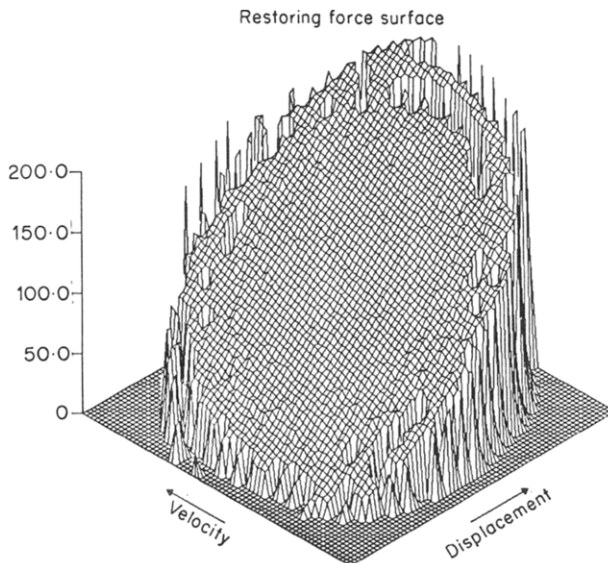


Figure 10. Force surface obtained using the estimated velocity and displacement data from the previous two figures.

Integration of a signal can be thought of as the solution of the simplest type of differential equation. This means that numerical routines for the integration of ordinary differential equations could be used. Bert and Stricklin [8] compare six methods, namely, the centred difference method, Runge-Kutta, Houbolt's method, Newmark's method, the Wilson theta method and the harmonic acceleration method. They test the methods on a number of second order systems. All these algorithms except the centred difference method are more complex and time consuming than those considered in this paper.

3.3. FREQUENCY CHARACTERISTICS OF INTEGRATION FORMULAE

It is now time to justify some of the remarks which have been made. Using ideas from Hammings book [9], one can obtain the frequency behaviour of the integration formulae

considered above. The simplest case is the trapezium rule. In this section, a time scale is used such that $\Delta t = 1$. This means that the sampling frequency is also 1, so the Nyquist frequency is 0.5, the angular Nyquist frequency is π . Using these conventions the trapezium rule (1) becomes

$$y_i = y_{i-1} + \frac{1}{2}(x_i + x_{i-1})$$

where x_i is the sampled input to the algorithm, and y_i is the integral. Using the backward difference operator $Z^{-1} = \exp(-d/dt)$, one can write the formula in the form

$$(1 - Z^{-1})y_i = \frac{1}{2}(1 + Z^{-1})x_i$$

so

$$y_i = \frac{1}{2} \cdot \frac{(1 + Z^{-1})}{(1 - Z^{-1})} x_i$$

so the pulse transfer function [10] for the integrator is

$$G(Z) = \frac{1}{2} \cdot \frac{(1 + Z^{-1})}{(1 - Z^{-1})}.$$

The more usual transfer function is obtained by simply substituting $Z = e^{i\omega}$. In this case

$$\begin{aligned} H(\omega) &= \frac{(1 + e^{i\omega})}{2(1 - e^{i\omega})} \\ &= \frac{\cos(\omega/2)}{2i \cdot \sin(\omega/2)}. \end{aligned}$$

Now, following Hamming, one can introduce a transfer function $Ha(\omega)$, which is a useful measure of the accuracy of the formula. It is defined as

$$Ha(\omega) = \frac{F(\text{estimated result})}{F(\text{true result})}.$$

Where F denotes the FT. Now, if $x = e^{i\omega t}$ the true integral is $e^{i\omega t}/i\omega$, and the estimated result is

$$\frac{\cos(\omega/2)}{2i \cdot \sin(\omega/2)} \cdot e^{i\omega t}$$

so $Ha(\omega)$ for the trapezium rule is given by

$$Ha(\omega) = \cos(\omega/2) \cdot \left(\frac{\omega/2}{\sin(\omega/2)} \right).$$

This is equal to unity at $\omega = 0$, and monotonically decreases to zero at $\omega = \pi$. This means that the trapezium rule only integrates constant signals without error. It underestimates the integral at all other frequencies.

Now, if one considers Simpson's rule (2) in the units of this section

$$y_{i+1} = y_{i-1} + (1/3)(x_{i+1} + 4x_i + x_{i-1})$$

or

$$(Z - Z^{-1})y_i = (1/3)(Z + 4 + Z^{-1})x_i$$

and the pulse transfer function is

$$G(Z) = \frac{Z + 4 + Z^{-1}}{3(Z - Z^{-1})}$$

Giving

$$H(\omega) = \frac{2 + \cos(\omega)}{3 \cdot \sin(\omega)}.$$

Probing with the signal $x = e^{i\omega t}$ yields the final result.

$$Ha(\omega) = \frac{2 + \cos(\omega)}{3(\sin(\omega)/\omega)}$$

and one can see easily that $Ha(\omega)$ tends to unity as ω tends to zero. This is the same behaviour as the trapezium rule. However, unlike the trapezium rule, $Ha(\omega)$ for Simpson's rule tends to infinity as ω approaches the Nyquist frequency, indicating instability at high frequencies. In Fig. 11, as in [9], $Ha(\omega)$ is shown for the three integration rules considered above. It shows clearly that Simpson's rule and Tick's rule becomes unreliable at high frequencies. There are in fact a whole family of Tick's rules, each is designed to make $Ha(\omega)$ flat over some region of the Nyquist interval. The one considered above is flat over the first half of the interval i.e. up to $f = 0.25$. The penalty one pays for the flat response is that $Ha(\omega)$ tends to infinity faster beyond some frequency.

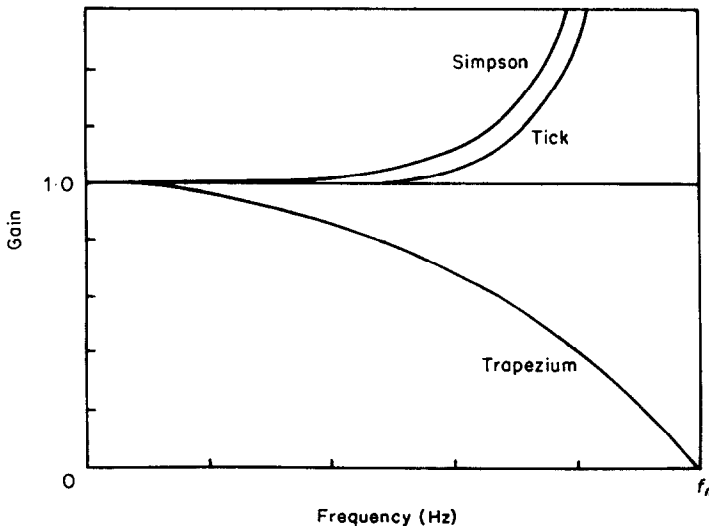


Figure 11. FRFs for various time domain integration procedures.

It still remains to show why these methods are unstable at low frequencies. The reason is that if one considers $Ha(\omega)$ for the trapezium rule it is only the whole story if x is known to an infinite precision. If the x_i have a measurement error or simply a truncation

error associated with them ζ_i , then the estimated integral is given by

$$\begin{aligned}\hat{y}_i &= G(Z) \cdot (x_i + \zeta_i) \\ &= y_i + G(Z) \cdot \zeta_i\end{aligned}$$

using the linearity of the procedure. So the error in the integral is given simply by

$$\varepsilon_i = \hat{y}_i - y_i = G(Z) \cdot \zeta_i.$$

So the transfer function $F(\text{output noise})/F(\text{input noise})$ is given by $\cot(\omega/2)/2i$ as above. This tends to infinity as ω tends to 0. This means that the trapezium rule greatly magnifies errors at low frequencies. It also shows that the integration is unstable under small perturbations. As the other two methods have the same $Ha(\omega)$ at low frequencies, they must suffer from the same problem.

In the examples considered above, the highest frequency of interest is 50 Hz when the band-limited input is used. The Nyquist frequency is 500 Hz. This gives a normalised value of 0.05 for the highest frequency of interest. One can see from Fig. 11 that up to this frequency the procedures are almost indistinguishable as regards accuracy. One is therefore justified in using the simplest rule to integrate. The trapezium rule offers a considerable saving in time if it can be used as one does not need to low-pass filter the output.

If frequencies are present in the signal up to say, one quarter of the Nyquist limit, one would probably need to use Tick's rule. At this upper limit, one can see from the diagram that $Ha(\omega)$ for the trapezium rule is less than 0.8, if one integrated twice one would only obtain 60% of the data at this frequency. If Simpson's rule were used, $Ha(\omega)$ is about 1.1 so integrating twice would give an overestimate by about 20%. One can see that Tick's rule gives a gain of unity right up to 0.25 just as it was designed to do. The transfer functions shown in Fig. 11 can be of considerable use in choosing the appropriate integrator for a problem.

4. FREQUENCY DOMAIN INTEGRATION

The theoretical basis of the FT method of integration is simple; if

$$Y_a(\omega) = \int_{-\infty}^{+\infty} dt e^{-i\omega t} \ddot{y}(t) = F\{\ddot{y}(t)\}$$

is the FT of the acceleration $\ddot{y}(t)$, then $Y_a(\omega)/i\omega = Y_v(\omega)$ is the FT of the velocity and $Y_a(\omega)/(-\omega^2) = Y_d(\omega)$ is the FT of the displacement. This means that in the frequency domain, division by $i\omega$ is equivalent to integration in the time domain. The mean removal can be carried out by simply setting the $\omega = 0$ line (one uses the digital FFT) to zero. Of course, one cannot divide by $i0$ anyway.

At first sight this seems like a very attractive way of looking at the problem, however it turns out to be beset by the same problems as the time domain methods and also to have a few of its own.

The first problem to arise concerns the nature of the acceleration signal. This need not be periodic and consequently the Fourier transform will have leakage problems [9]. One effect of leakage can be the introduction of low frequency trends into the acceleration signal which will be magnified by the integration. Figure 12 shows the FT of (a) a sine wave which was periodic over the number of points taken and (b) of a sine wave which was not. One can see that energy has "leaked" out to higher and lower frequencies. It is the low frequencies that cause problems for integration. To avoid this one multiplies the

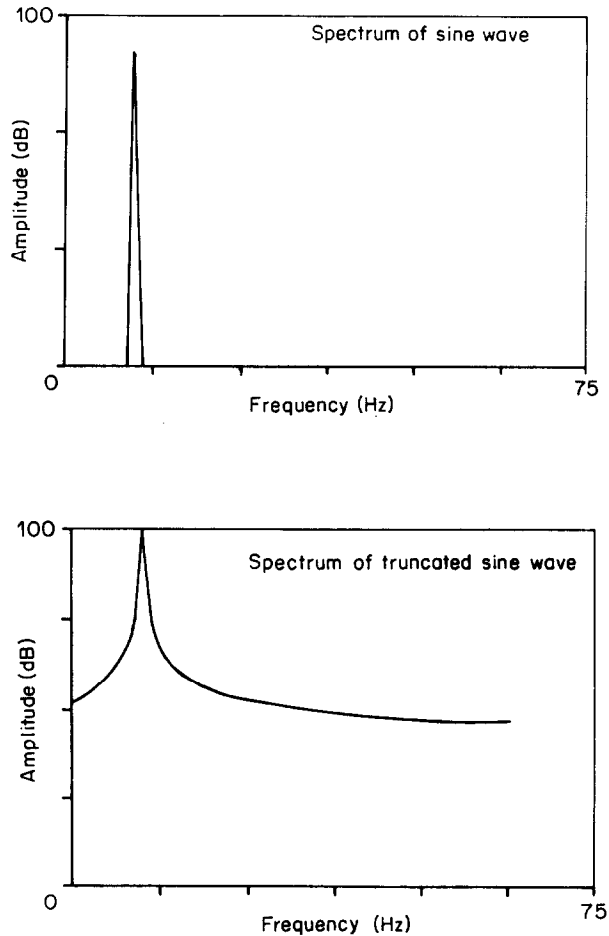


Figure 12. The effect of leakage on the spectrum of a sine wave.

N points of data to be transformed, by a window which makes the data periodic over the record. The window used in this study is the Hanning window [9]. The Hanning window is only close to unity in the centre so only velocity and displacement data obtained from the centre of the window are reliable. To process the whole acceleration record one steps through the data using a series of overlapping FTs. The loss factor used below is simply the percentage of data discarded from each FT. The effect of multiplying by the windows is the introduction of an amplitude modulation into the signal, this can be made negligible by taking a high enough loss factor. Unfortunately one may need many FTs. It is possible that other windows may be more efficient.

Example 6. First, the system was excited with a sine-wave periodic over each 32 sample points. The acceleration obtained was then integrated twice by the method above using a 1024 point FT with no windowing. There were no leakage effects and the integration was very accurate, the MSE for the comparison with the exact data was 0.562 and the comparison is shown in Fig. 13. Next, the same data was integrated using a Hanning window and a loss factor of 80%. The comparison with the exact data is shown in Fig. 14. The MSE of 0.854 is still excellent. The only difference between the two procedures is that the second produced a small amplitude modulation. The acceleration was then obtained for a sine excitation periodic over 31 sample points. When the integration was

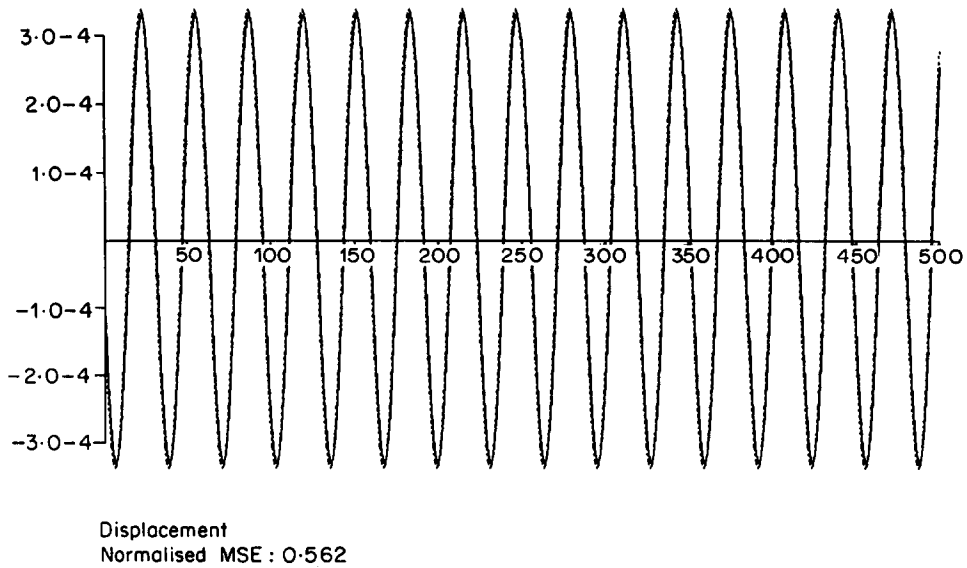


Figure 13. Comparison of the exact and estimated displacement data when the system is excited by a sine wave which is periodic over the FFT rectangular window. ---, Measured output; —, estimated output.

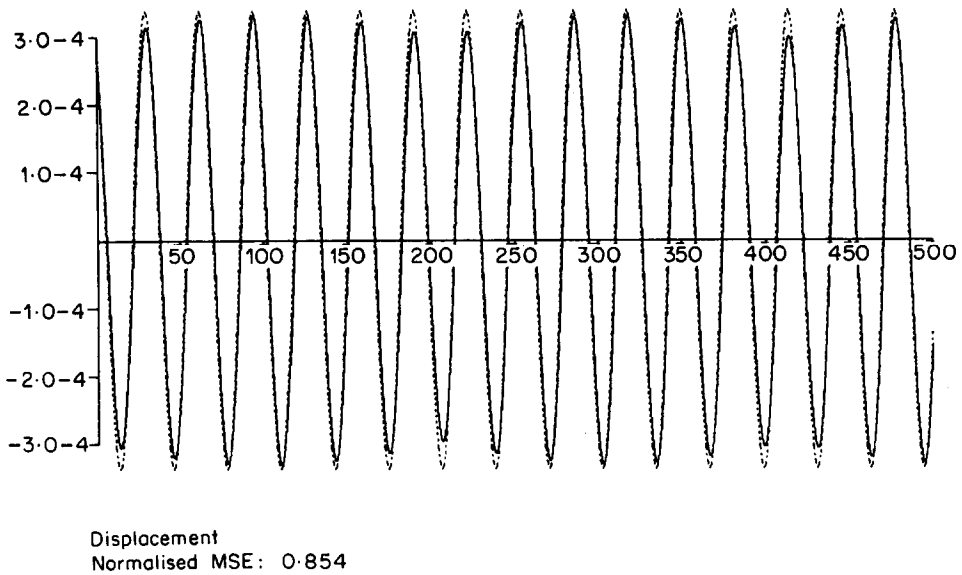


Figure 14. Comparison of the exact and estimated displacement data when the excitation is a sine wave periodic over the FFT window. Hanning window used. ---, Measured output; —, estimated output.

carried out as before without windowing, the results were as shown in Fig. 15. A large low frequency component has been introduced, the very high MSE of 344.0 for the comparison reflects this. The integration was then carried out using the window and the same loss factor as before. The results are now considerably improved. The comparison MSE is 0.909. The comparison is shown in Fig. 16. The results are indistinguishable from those in Fig. 14. This example shows that the leakage problem is reduced by windowing.

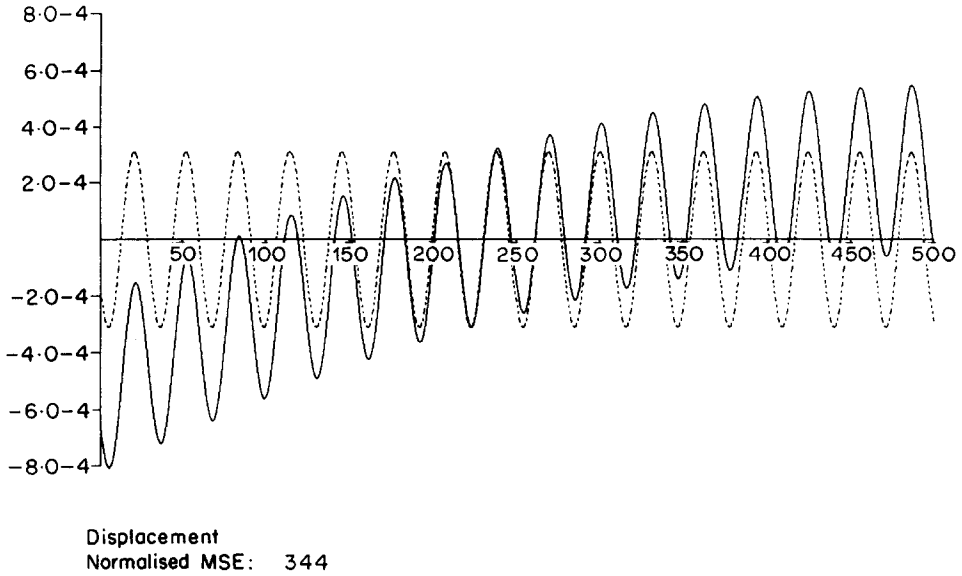


Figure 15. Comparison of the exact and estimated displacement data when the excitation is a sine wave which is not periodic over the FFT rectangular window. ---, Measured output; —, estimated output.

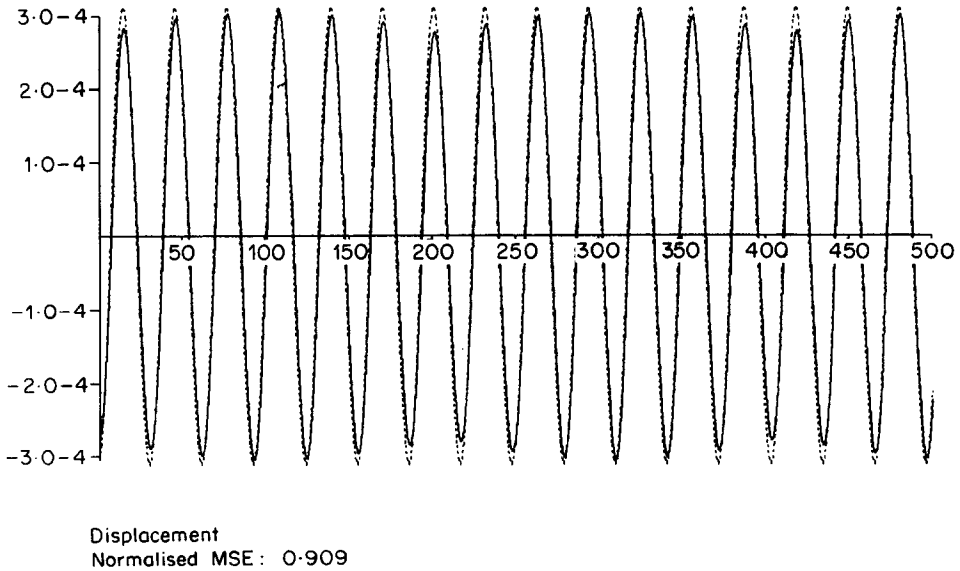


Figure 16. Comparison of the exact and estimated displacement data when the excitation is a sine wave which is not periodic over the FFT window. Hanning window used. ---, Measured output; —, estimated output.

The next and most serious problem is the magnification of low frequency trends in the data by the integration procedure.

Example 7. Data was simulated for the system as in example 1. The acceleration data was then integrated using a 1024 point FFT. A Hanning window and a loss factor of 80%

were used. The $\omega = 0$ line was deleted. The resulting errors for the velocity and displacement data were $\text{MSE}(\dot{y}) = 1.01$ and $\text{MSE}(y) = 3.51$. A parameter estimation yielded

Coefficients	% error
$\hat{m} = 1.002$	0.2
$\hat{c} = 42.4$	6.0
$\hat{k} = 1.012 \times 10^4$	1.2
$\hat{\alpha} = 0.0$	0.0

with a model MSE of 3.85%. The results compare favourably with the time domain methods, however, the FT method is considerably more time consuming.

Table 3 shows the displacement and velocity error if one uses various different loss factors and numbers of lines for the FFT.

TABLE 3
Mean-square errors for the integrated velocity and displacements

No. of lines	Loss factor (%)						
	50	60	70	80	90	95	100
128	43.783	34.065	31.602	29.792	28.472	28.276	28.171
	10.276	6.278	3.909	2.708	2.465	2.395	2.376
256	17.307	13.832	12.512	11.840	10.337	10.226	10.191
	7.316	4.040	2.129	1.119	0.861	0.821	0.816
512	10.335	8.108	5.825	4.750	4.645	4.559	4.544
	6.543	3.605	1.888	1.009	0.732	0.702	0.694
1024	9.577	6.818	4.539	3.513	3.639	3.552	3.563
	6.648	3.648	1.738	1.006	0.721	0.687	0.685

The FT method of integration was used with various different loss factors and numbers of lines in the FFT. The top number in each pair is $\text{MSE}(y)$ and the other is $\text{MSE}(\dot{y})$.

The low frequency component appears to arise because in integrating the data twice one had to divide by $-\omega^2$. This means that any noise in the data at low frequencies will be magnified by a factor ω^{-2} . Thus errors at $\omega = 0$ would be infinite. There will almost always be some noise at low frequencies as aliasing will fold back energy from high frequency noise into this region to add to any noise already present. This is of course the frequency-domain analogue of the time-domain noise magnification mechanism discussed earlier.

The way around this problem, used successfully by Al-Hadid and Wright [12] is to choose a cut-off ω_c by integrating the data and looking at the *averaged* transform of the displacement, one chooses the cut-off above any spurious low frequency peaks. The integration is then repeated except this time the value of $Y(\omega)$ for each $\omega < \omega_c$ is replaced in each FT by the value on a line segment joining $Y(0)$ to $\bar{Y}(\omega_c)$ the averaged value of Y at ω_c .

A much simpler way of dealing with this problem is to stop it arising in the first place.

Example 8. Data was simulated as in the previous example except that $x(t)$ was filtered to give a band-limited input 5-30 Hz. Again a 1024 point FFT was used with an 80% loss

factor. The velocity and displacement errors were respectively 0.44 and 0.52. Parameter estimation gave the model

Coefficients	% error
$\hat{m} = 1.005$	0.05
$\hat{c} = 40.9$	2.25
$\hat{k} = 1.06 \times 10^4$	6.00
$\hat{\alpha} = 0.00$	0.00

with a model MSE of 0.34%. The force surface obtained is shown in Fig. 17. The linearity of the system is evident. Note that the division by ω^2 means that the FT method does not suffer from high frequency problems.

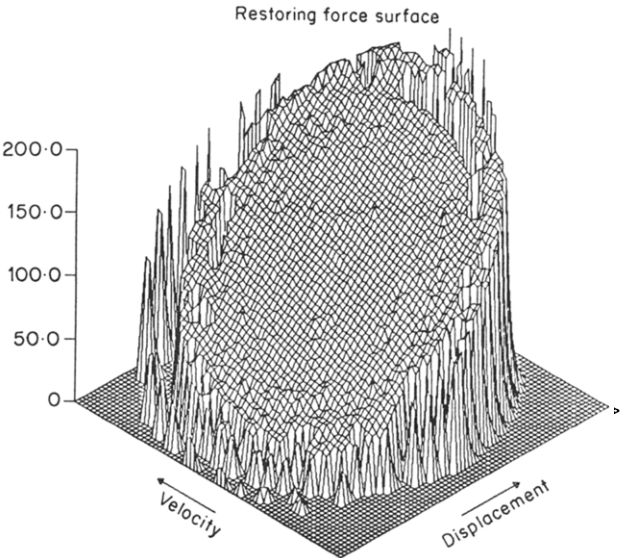


Figure 17. Force surface obtained using velocity and displacement data from frequency domain integration of acceleration data. The system is excited with a band-limited signal.

5. INITIAL CONDITIONS

It is important to mention a point concerning initial conditions and integration constants. It is not always appropriate to fix constants of integration by removing the mean level. If one considers a system with a symmetric (i.e. even) non-linearity, then even if the input is zero-mean the system will generate a mean level as an essential part of the output. The way around this problem is to modify one's experimental procedure. At the same time as one records the acceleration data one can take a sample of displacement data in order to estimate the mean level, the integrated data can then be adjusted to have this mean level. A much simpler procedure is to start recording data with the system at rest before the input force is applied. This fixes the initial conditions as $y(0) = 0, \dot{y}(0) = 0$. After the integration the null data can be deleted.

6. DIFFERENTIATION OF MEASURED TIME DATA

Because differentiation is defined as a limiting process it is difficult to carry out numerically. The approximation

$$\frac{dy}{dt} = \lim_{\delta t \rightarrow 0} \frac{\delta y}{\delta t} \approx \frac{\Delta y}{\Delta t} \quad (\Delta y, \Delta t \text{ finite})$$

will clearly become better as one takes smaller and smaller Δt . Unfortunately this is the sort of operation which will produce significant errors from round-off when one tries to perform the calculation on a computer (unless one has accuracy to an arbitrary number of decimal places as in ADA, even so the equipment used for the data capture will have limited accuracy). Thus numerical differentiation requires one to walk a tightrope between approximation errors and round-off errors. For this reason it is not recommended that one should measure $y(t)$ and numerically differentiate to form $\dot{y}(t)$ and $\ddot{y}(t)$ for the force surface problem. Unfortunately there are some situations where one has no choice. For example if one applied the method to a rotor system in order to estimate bearing coefficients, one would not easily be able to measure the vertical and horizontal accelerations of the rotor, one usually uses a non-contact displacement transducer in rotor dynamic problems. For this reason some methods of numerical differentiation are considered.

7. TIME DOMAIN DIFFERENTIATION

7.1. CENTRED DIFFERENCE FORMULAE

These formulae implement the differentiation as a digital filter or recursion relation, the 3-, 5- and 7- point centred difference formulae are given by

$$\dot{y}_i \approx (y_{i+1} - y_{i-1}) / 2\Delta t \quad (4)$$

$$\dot{y}_i = [-y_{i+2} + y_{i-2} + 8(y_{i+1} - y_{i-1})] / (12\Delta t) \quad (5)$$

$$\dot{y}_i = [2(y_{i+3} - y_{i-3}) - 13(y_{i+2} - y_{i-2}) + 50(y_{i+1} - y_{i-1})] / (60\Delta t). \quad (6)$$

It is important to monitor the remainder terms when using these formulae. A careful analysis including them can indicate why one is not guaranteed greater accuracy if one uses a higher order formulae.

Example 9. The system used in the previous examples was simulated with input $x(t)$ a Gaussian sequence band-limited in the range 0–200 Hz. The resulting displacement data was then differentiated twice using a 5-point centred difference. The resulting velocity and acceleration errors were 0.015 and 7.84 respectively. Again the velocity estimate is almost perfect. A least-squares parameter estimation gave

Coefficients	% errors
$\hat{m} = 0.961$	3.90
$\hat{c} = 55.2$	38.0
$\hat{k} = 0.98 \times 10^4$	2.0
$\hat{\alpha} = 0.0$	0.0

with a model MSE of 19.5%. The force surface obtained is shown in Fig. 18. The reason why the model is so much in error even though the parameters are quite good is because the error in $\ddot{y}(t)$ appears as a time-shift which decorrelates $x(t)$ and $\ddot{y}(t)$ (Fig. 19). This is reflected in the parameter estimate, inaccuracies in the damping generate phase errors

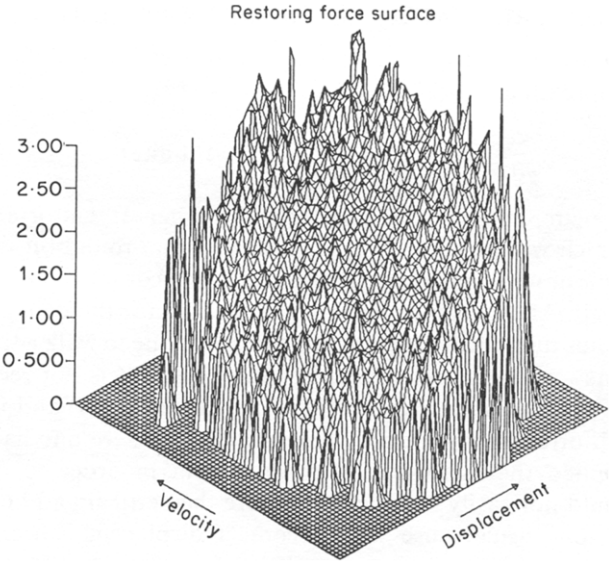


Figure 18. Force surface obtained using velocity and acceleration data from a 5-point centred difference formula.

or time-lags in the output. The reason for this shift is not obvious. One might suggest that the algorithm, considered as a digital filter is not zero-phase and one could solve the problem by taking the derivatives in both the forward and backward directions and then averaging the result. However, this does not appear to help, in the case of this example anyway. Because one can envisage situations where one would not measure acceleration data, for example, if one is studying the movement of a rotor one, further study of numerical differentiation is indicated.

The results of using the various centred difference formulae on the data of example 9 are shown in Table 4.

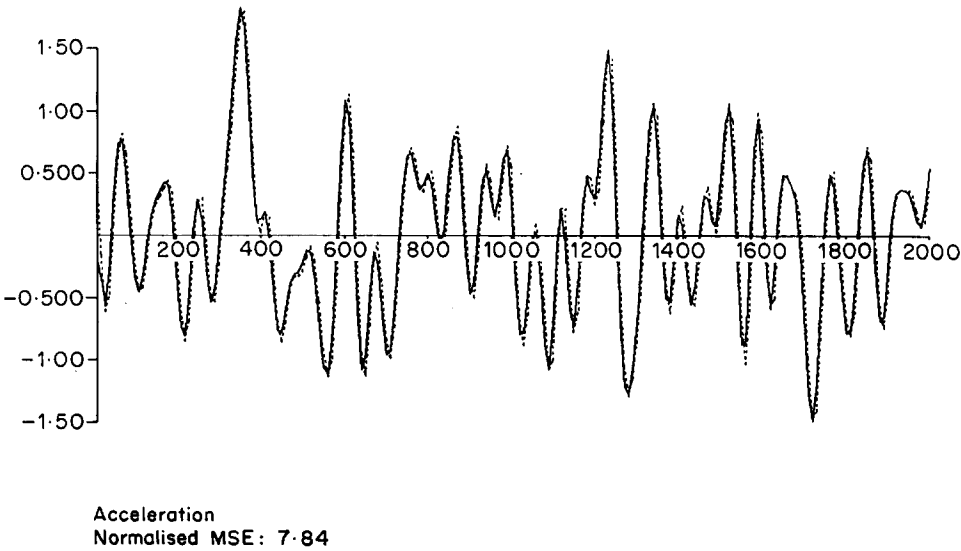


Figure 19. Comparison of exact and estimated acceleration data obtained by using the 5-point centred difference formula twice on displacement data. ---, Measured output; —, estimated output.

TABLE 4

Mean-square errors for the differentiated time data

	MSE ($\dot{y}$)	MSE ($\ddot{y}$)
3-point	0.017	8.600
5-point	0.016	7.836
7-point	0.007	8.213

Centred difference formulae were used.

TABLE 5

Mean-square errors for the estimated derivatives

Order	MSE ($\dot{y}$)	MSE ($\ddot{y}$)
2	0.138	9.080
4	0.015	9.836
6	0.012	8.093
8	0.014	8.185

Polynomial fitting was used.

As with the integration formulae one can obtain the frequency characteristics of each of the formulae above by using the techniques of section 3.3.

7.2. DIFFERENTIATING FITTED POLYNOMIALS

This method is very simple in principle. One fits a polynomial to N data points (N odd) such that the point at which the derivative is required is at the centre. One then takes the analytic derivative of the fitted polynomial. This is rather time consuming and for low orders of polynomial simply reproduces the results of using centred differences (slight deviations arise due to the way the algorithms are programmed). The results of differentiating the data of example 9 using various polynomial orders are shown in Table 5.

If one were to use this method, one would have to be sure that the data is sampled correctly. If it is undersampled the situation shown in Fig. 20 might arise.

8. FREQUENCY DOMAIN DIFFERENTIATION

The basis of this method is exactly the same as for integration except that to differentiate one multiplies the FT by $i\omega$ instead of dividing, i.e. $Y_v(\omega) = i\omega Y_d(\omega)$. It is obvious from the frequency-domain argument that differentiation greatly magnifies high frequency noise.

The leakage problem is exactly the same as for integration and is dealt with in exactly the same way. Table 6 shows the errors in the differentiated signals as a function of loss factor (Hanning window used) and number of FFT lines.

If one differentiates the data of the previous example using this method one still observes the phase drift in the acceleration data obtained. The method in terms of the continuous FT is zero-phase. In fact $Ha(\omega)$ is 1. (This is true if one disregards the effects of windowing etc. If one takes account of these effects correctly one would arrive at a more complicated form for $Ha(\omega)$.) Thus, no simple explanation of the time delay arises from considering the problem in the frequency domain.

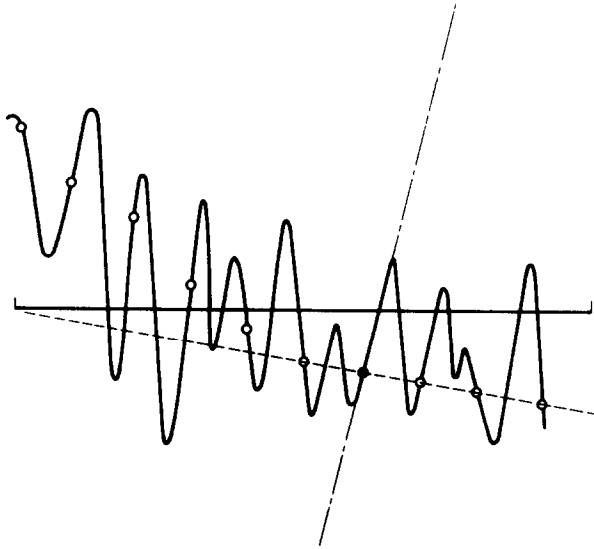


Figure 20. The effect of undersampling on the estimation of derivatives. —, True tangent; ----, estimated tangent; ⊗, sample point.

TABLE 6
Mean-square errors for the velocity and acceleration estimates

No. of lines	Loss factor (%)						
	50	60	70	80	90	95	100
128	9.771	5.884	3.306	1.660	0.691	0.508	0.471
	16.177	13.124	11.237	10.048	9.392	9.469	9.410
256	6.903	4.134	2.316	1.162	0.812	0.730	0.714
	14.040	11.737	10.409	9.411	9.492	9.356	9.394
512	6.357	3.769	4.223	2.402	2.537	3.088	2.550
	13.645	11.328	11.500	10.291	10.722	10.964	10.760
1024	10.187	8.890	5.045	4.940	4.648	4.695	4.702
	17.449	16.511	12.743	13.594	13.229	13.412	13.361

The FT method of differentiation was used with various different loss factors and numbers of lines in the FFT. The top number in each pair is the velocity error.

To conclude this section then, numerical differentiation of time data clearly has problems. For this reason it is suggested that one should always measure acceleration where possible and integrate to obtain velocity and displacement.

9. SUMMARY

It is shown that numerical integration procedures can be used to reliably produce estimated velocity and displacement data from acceleration measurements by a variety of methods. Numerical differentiation is shown to be less reliable. The constants of integration can only be fixed in some cases by using a particular experimental procedure, this leads one to consider whether appropriate design of experiments can reduce or remove processing problems. Experimental strategy is the subject of the second part of this work [13].

The conclusions regarding integration and parameter estimation extend trivially from sdof systems to mdof systems. No new problems arise except that the data from each measurement point must be sampled simultaneously.

One would hope that the procedures above retain their usefulness when experimental data is considered rather than simulated. An experimental study of a number of sdof-like systems has been carried out [14]; the results indicate that the integration procedures appear to be insensitive to measurement noise.

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